Do Now · retrieval

Lesson 144 · DNS and IP Routing: From Name to Destination

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Which TCP/IP layer handles routing and IP addressing?

Q2

What data structure uses LIFO?

Q3

What is a trace table used for?

Q4

How many bits are in one byte?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Resolve hostname — Browser requests address for a website name through DNS.

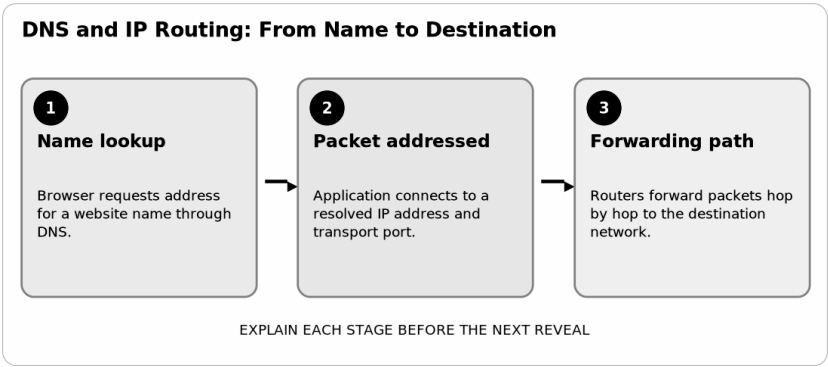
Core idea 2

Use IP destination — Application connects to a resolved IP address and transport port.

Core idea 3

Route across networks — Routers forward packets hop by hop to the destination network.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

DNS — Hierarchical name-resolution system.

IP address — Network-layer address used to route packets.

resolver — Client/service that queries DNS and caches answers.

router — Forwards IP packets towards destination networks.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

A pupil types example.org and receives an IP address from DNS. Explain what DNS does and does not do.

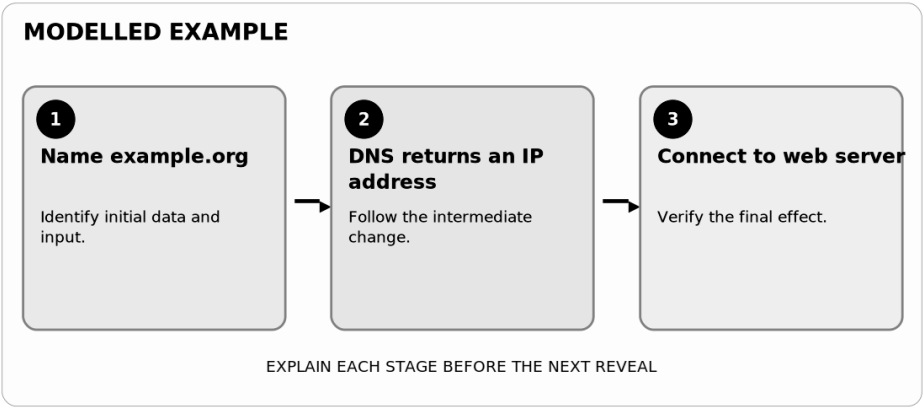
Follow the worked method

1. Name example.org — Identify the relevant information and make a prediction.

2. DNS returns an IP address — Explain the change using correct technical terms.

3. Connect to web server — DNS maps the hostname to relevant resource records such as an A/AAAA address. DNS does not directly transmit the web page. Browser makes a separate connection to the chosen server address and requests content, often over HTTPS.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Cached old address
Identify the relevant information and make a prediction.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Explain how DNS helps a user retrieve a website by name. [6]

Question 2

Differentiate DNS name resolution from IP packet routing. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

A school website moves servers but some users reach the old address. Explain cached DNS and propose sensible checks. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

DNS sends the entire web page to the browser after it receives the hostname.

Use key vocabulary from memory

1. Explain DNS in your own words.

2. Explain IP address in your own words.

3. Explain resolver in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.